

# How Far Can General-Purpose LLMs Go in Tigrinya? A Lexicon-Grounded Evaluation with Trivial Baselines

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## Abstract

We evaluate eight general-purpose large language models on three lexicon-grounded Tigrinya tasks — part-of-speech tagging, morphosyntactic labelling, and English→Tigrinya translation — using a 5,783-row annotated bilingual lexicon. Our central methodological finding is that such evaluations are easy to over-read: a permissive matching rule puts the best model at 85.9% on part-of-speech tagging and 89.4% on morphosyntactic labelling, but scoring the same saved outputs by the first word the model commits to costs up to 26 points, and a constant-answer baseline reaches 53.7% while an English-only tagger that never sees the Tigrinya reaches 60.4%. Measured against those baselines, three of eight models demonstrate above-trivial POS performance and two exceed the English-only baseline on morphosyntactic labelling, by under three points. On translation all eight models score at or below chrF 0.81, while NLLB-200-3.3B, which supports Tigrinya, reaches chrF 16.05 on identical items with an identical scorer. We argue that low-resource evaluations without trivial baselines are uninterpretable, and release the lexicon, the scoring code and the saved model outputs. Because the procedure we apply here — report every matching rule, measure against baselines that use none of the target language, and quantify prompt echo — requires no resources beyond those an evaluation already has, it transfers directly to other low-resource languages, where the same scoring choices determine whether a reported number reflects competence or an artefact of measurement.

## 1 Introduction

Tigrinya is a Semitic language of Eritrea and northern Ethiopia with approximately 9–10 million speakers and very little machine-readable linguistic infrastructure. This scarcity makes it a natural test of whether general-purpose LLMs generalise beyond high-resource languages, and a difficult one to

measure: without established benchmarks it is easy to produce numbers that look like competence.

This paper makes three contributions.

1. **A lexicon-grounded benchmark** of 5,783 English–Tigrinya rows (4,331 English headwords, 3,767 Tigrinya forms), each carrying a part-of-speech label, derived from a digitised bilingual dictionary (Union of International Democrats, 2015).
2. **An evaluation of eight LLMs** across POS tagging, morphosyntactic labelling and translation, reported under three matching rules and against two trivial baselines.
3. **A methodological result:** on a lexicon-scale task, the choice of matching rule moves reported accuracy by up to 26 points, and two baselines that use no Tigrinya at all reach 53.7% and 60.4%. Papers in this setting that report neither are not interpretable.

The third contribution is the reason for this paper. An earlier version of this work reported results computed with a permissive matching rule and no baselines, and drew an architectural conclusion that does not survive either correction. The authors have requested its retraction; Section 7 states the relationship between the two.

## 2 Related work

### 2.1 Tigrinya and Ge’ez-script NLP

Tigrinya remains severely underrepresented in NLP: a recent survey of over 50 studies from 2011 to 2025 finds a field whose progress is driven almost entirely by discrete resource-creation milestones rather than by modelling advances (Gaim and Park, 2026), a pattern echoed in broader surveys of Ethiopian-language NLP (Tonja et al., 2023). The resources that exist are small and task-specific. Öktem et al. (2020) built a domain-specific Tigrinya→English

NMT system using transfer learning from other Ge’ez-script languages, reporting a 1.3 BLEU improvement over a neural baseline; [Tela et al. \(2020\)](#) showed that an English XLNet fine-tuned on 10k examples reaches 78.88 F1 on Tigrinya sentiment, outperforming mBERT by 7 points; TiGQA contributed 2,685 expert-annotated question–answer pairs from 537 context paragraphs ([Teklehaymanot et al., 2024](#)); and VEXMLM demonstrated that vocabulary extension with 30k Ge’ez-script subwords substantially improves XLM-R on Amharic and Tigrinya, attributing the baseline weakness to high out-of-vocabulary rates under Latin-centric tokenizers ([Teklehaymanot et al., 2026](#)). Tigrinya also appears as one slice of multilingual benchmarks — MasakhaPOS, the largest POS dataset for African languages, covering 20 typologically diverse languages under UD guidelines ([Dione et al., 2023](#)), and Belebele, a parallel multiple-choice reading-comprehension set in 122 language variants ([Bandedkar et al., 2024](#)). Our lexicon complements these: it is not running text, and it tests the dictionary-headword level that sits beneath all of them. The structural reason such resources are scarce at all is the one [Joshi et al. \(2020\)](#) documented — representation in NLP venues tracks resource availability, calling into question the “language agnostic” status of current systems.

## 2.2 LLM evaluation for low-resource languages

NLLB-200 added Tigrinya (tir\_Ethi, Ge’ez script, Afro-Asiatic/Semitic) to a 200-language translation model ([NLLB Team et al., 2022](#)), evaluated on FLORES, whose professionally translated, multilingually aligned design set the standard for low-resource MT benchmarking ([Goyal et al., 2022](#)). Against that backdrop, several studies have established that general-purpose LLMs lag purpose-built MT on exactly the languages we study. [Robinson et al. \(2023\)](#) evaluated ChatGPT on 204 FLORES-200 varieties and found NLLB superior on 169 of 201 languages (84.1%), by an average of 11.9 chrF, with the gap widest for African languages and the Ge’ez script specifically; they further report that five-shot prompting improved on zero-shot by only 0.88 chrF on average across 203 directions, which is consistent with our own few-shot probe moving gemma-7b from chrF 0.16 to 0.66 with exact match unchanged. [Hendy et al. \(2023\)](#) reach the same conclusion for GPT models across eighteen directions — competitive for high-resource, limited

for low-resource — and [Zhu et al. \(2023\)](#) find GPT-4 beating NLLB in 40.91% of directions overall while still trailing badly on the long tail. [Adebara et al. \(2025\)](#) extend this to a continent-scale African benchmark and attribute the residual disparities to data inequities rather than model scale. Our Section 5.3 result — three NLLB checkpoints at chrF 14.1–16.1 against eight LLMs at  $\leq 0.81$  — is this finding at lexicon granularity, where the LLM floor is not merely lower but absolute.

A separate strand asks whether LLM evaluations in these settings are conducted carefully enough. LArABench benchmarked 33 Arabic tasks across 61 datasets and explicitly computed a random baseline “to determine if the LLMs predictions are not merely the result of chance” ([Abdelali et al., 2023](#)). We regard that step as mandatory rather than optional, and Section 5 shows why.

## 2.3 Trivial baselines and spurious cues

The methodological core of this paper has a direct precedent in natural language inference. [Poliak et al. \(2018\)](#) showed that a hypothesis-only model — one that never sees the premise, a degenerate solution by construction — significantly outperforms a majority-class baseline on a number of NLI datasets; [Gururangan et al. \(2018\)](#) showed the same artefacts let a simple text classifier label the hypothesis alone correctly in about 67% of SNLI and 53% of MultiNLI, concluding that model success “has been overestimated”. [McCoy et al. \(2019\)](#) demonstrated that strong test-set scores can rest on fallible syntactic heuristics that collapse on controlled sets, and [Niven and Kao \(2019\)](#) found BERT’s 77% on argument reasoning comprehension to be entirely accounted for by spurious statistical cues. Our English-only tagger is a hypothesis-only baseline in this sense: the prompt format `Phrase: {Tigrinya} ({English})` leaves the part of speech largely recoverable from the gloss, and the baseline reaching 60.38 is the lexicon-task analogue of the 67% SNLI result.

For POS tagging specifically, [Kann et al. \(2020\)](#) make the closest argument to ours: weakly supervised taggers reported to perform “almost as well as supervised ones” were evaluated on languages unlike truly low-resource ones, and on 15 genuinely low-resource languages the best model gets fewer than half the words right, with a strongest baseline macro-average of 39.11%. They also note that such methods presuppose “high-coverage and almost error-free dictionaries” — precisely the assumption

our Section 3.3 annotator notes undermine, where the source dictionary contains mislabelled parts of speech and outright mistranslations. At the level of benchmark design, Reuel et al.’s assessment of 46 best practices across 24 benchmarks identifies stating random performance as a scored criterion, since without it one cannot separate capability from chance or metric design (Reuel et al., 2024).

## 2.4 Scoring rules, answer extraction and prompt artefacts

Our finding that the matching rule moves reported accuracy by up to 26 points on fixed outputs belongs to a growing literature on evaluation mechanics. Tam et al. (2024) show that constraining LLMs to structured output formats significantly degrades measured reasoning, and disentangle format errors from content by using an LLM as a “perfect parser” rather than a regex — the mirror image of our problem, where permissive parsing manufactures competence instead of destroying it. Kirouane and Petrocheilos (2026) document the same failure mode concretely in a low-resource setting: a scorer that scanned the whole response instead of the requested answer line produced a +29.8 pp artefact, one-directional against verbose outputs, and they conclude that “any benchmark that requests an answer format must score that format first and report how often it was absent”. Our first-token rule is that recommendation applied; our exact / first-token / set-intersection triple reports the spread it would otherwise hide.

The prompt-echo column in Section 5.2 relates to known label-copying behaviour. Zhao et al. (2021) show that few-shot LLM accuracy is destabilised by bias towards answers appearing in the prompt or common in pretraining, varying “from near chance to near state-of-the-art” with prompt format alone. When the gold distribution is 53.7% noun and the prompt example contains the word noun, a model that copies the example is scored correct by construction under set-intersection — gemma-7b does this on 97.4% of answers. We therefore treat echo rate as a reportable diagnostic rather than a footnote.

## 2.5 Metrics and agreement statistics

We report chrF (Popović, 2015) because character  $n$ -gram F-score is tokenisation-independent and therefore appropriate for Ge’ez-script targets averaging 1.37 whitespace tokens, where word-level BLEU (Papineni et al., 2002) is near-undefined; we follow Post’s argument that BLEU is a param-

eterised family whose configurations are not comparable across papers unless the scheme is pinned down (Post, 2018), and report character-level BLEU alongside chrF for this reason. Agreement is reported as Cohen’s  $\kappa$  (Cohen, 1960). We note the standard caveat that  $\kappa$  is hard to interpret under class imbalance — the “kappa paradox” — and that absolute thresholds are not meaningful without a contextual reference (Wong et al., 2021); this is why Section 3.3 reports  $\kappa$  beside raw observed agreement and beside the filled-label-only variant, rather than reporting a single coefficient.

## 2.6 Dictionaries as evaluation resources

Digitised bilingual dictionaries are a standard bottom-up resource for languages lacking parallel corpora: Wickramasinghe and de Silva (2023) argue that for low-resource pairs it is “more feasible to move in the bottom-up direction where finer granular pairs such as dictionary datasets are developed first”, and Alnajjar et al.’s Ve’rdd addresses the complementary problem of grassroots paper dictionaries edited by many hands, which require re-evaluation rather than direct ingestion (Alnajjar et al., 2020). Our source exhibits exactly the defects that motivate that work — a direction-swapped second block, inline gender markers embedded in the target field, and annotator-flagged mistranslations — and Sections 3.1–3.3 document them rather than silently cleaning them.

## 2.7 Relation to the earlier version

An earlier version of this work introduced the lexicon and the four-task framework used here. Section 7 states what it reported, which of its claims survive re-scoring, and the authors’ request for its retraction.

# 3 Dataset

## 3.1 Source and construction

The lexicon was digitised from the Swansea Tigrinya–English dictionary (Union of International Democrats, 2015), with additional entries contributed by native-speaker linguists, and carries the source dictionary’s own part-of-speech abbreviations. The digitised source holds 7,234 rows concatenated from two directions, English→Tigrinya followed by Tigrinya→English, under a single pair of column headers; in the second block the column named English therefore holds Ge’ez script. We detect that block by script, reorient it, and merge

both directions, keeping distinct senses of one form as separate items — *estimate* → ጥምት is retained as both noun and verb — which yields **5,783 gold rows** over 4,331 English headwords and 3,767 Tigrinya forms. Three evaluation set sizes follow from this and are not interchangeable: 5,783 gold rows, **5,775 items** for translation and lexical alignment after each task drops duplicate pairs and rows with an empty side, and **5,701 items** for the two tagging tasks, which were run against a further reduction representing each multi-sense entry by its first sense. Appendix A gives the row-level derivation, the provenance of the source file, and the reconstruction of the 5,701-item set from the released gold.

### 3.2 What the annotations do and do not contain

The POS labels are the source dictionary’s, not independently re-annotated by us. They are coarse: noun accounts for 3,109 of 5,783 labelled entries (53.7%), verb 1,542, adjective 879.

Morphosyntactic features are **sparse**. Only 185 entries carry gender, number or plurality; the remaining 5,598 (96.8%) carry a value identical to the POS label. We therefore do not treat morphosyntactic labelling as a distinct linguistic dimension, and report it in Section 5.2 as what it measures: a second POS task with a different prompt.

### 3.3 Inter-annotator agreement

A 500-item subset of the lexicon was double-annotated for part of speech, gender, number, category agreement and lexical alignment. The records are released as POS\_english\_to\_tigrigna\_Annotated.xlsx — two annotator sheets, an adjudicated sheet and the annotation guide that defines each value — and scripts/compute\_iaa.py recomputes every figure below from them.

An annotator sometimes left a cell empty where the other assigned a label. Those are not matching judgements, so we count them as disagreements and report Cohen’s  $\kappa$  over all 500 items, with a blank treated as its own category (Table 1).

Restricting instead to the items both annotators labelled raises every dimension — 0.9076, 1.0000, 1.0000, 0.9815, 0.9850 on 500, 420, 428, 445 and 474 items. We report the stricter figures because the difference between the two columns is largely a difference in what each annotator left blank: Annotator 1 omits 32 gender, 32 number and 46 agreement labels that Annotator 2 supplies, against 6, 4

| Dimension          | $\kappa$ | Observed |
|--------------------|----------|----------|
| Part of speech     | 0.9076   | 0.9500   |
| Gender             | 0.8079   | 0.9240   |
| Number             | 0.6762   | 0.9280   |
| Category agreement | 0.5912   | 0.9000   |
| Lexical alignment  | 0.8487   | 0.9480   |

Table 1: Inter-annotator agreement on the 500-item subset, blanks counted as disagreements. All five dimensions are computed over all 500 items.

and 3 in the other direction. Discarding those rows would hide the main source of divergence.

Three properties of this subset bound what the coefficients support.

**The two columns are not independent of the adjudication.** On the 25 part-of-speech items where the annotators differ, the adjudicated sheet reproduces Annotator 2 on all 25 and Annotator 1 on none; on the three gender, agreement and alignment disagreements it reproduces Annotator 1. The adjudication is assembled from the two columns rather than decided separately, so it cannot serve as an independent third judgement.

**Agreement on filled labels is near-total.** Where both annotators assigned a gender or number value, they never differ — 0 disagreements across 420 and 428 items. The divergence between them is entirely in coverage, not in choice of label, which is why  $\kappa$  falls as far as it does once blanks count.

**The subset is not a random sample.** Tracing each item to its source sheet shows the 500 are the alphabetical head of each category: all 9 prepositions, all 10 pronouns and all 6 interjections, then 305 adjectives and 150 nouns from the tops of those sheets. The corpus is 54% noun and 25% verb; this subset is 61% adjective and contains **no verbs**. The coefficients therefore describe agreement on this subset and are not an estimate for the corpus. Establishing one requires a stratified re-annotation covering verbs, which we leave to future work (Section 9).

The workbook also carries the annotators’ free-text notes on 201 of the 500 items, documenting mislabelled parts of speech in the source dictionary, inflected forms given as citation forms, and outright mistranslations (*at least* → ብብዚኹ, which means *at most*). These support the caveat in Section 3.2 that the POS labels are the source dictionary’s and were

not independently verified.

## 4 Experimental setup

Eight models, all loaded through the Hugging Face transformers interface (Wolf et al., 2020) and run on a single compute node with four A100 GPUs, 32 CPU cores and 128 GB of RAM (Table 2).

**We do not compare architectures.** All five causal models are instruction-tuned and all three sequence-to-sequence models are raw pretrained checkpoints; the two factors are perfectly confounded in this model set and no result here can separate them. The three base seq2seq models receive span-infilling prompts matched to their pre-training objective; they nonetheless produce almost no usable output, which we report as a property of those checkpoints on this task rather than of their architecture.

Generation is greedy with `max_new_tokens=128`, `min_new_tokens=1`. xlm-roberta-base (Conneau et al., 2020) was included initially but fails on all tasks — its prompts contain no `<mask>` token — and is excluded.

### 4.1 Scoring

Model answers are frequently verbose: gemma-7b’s median POS answer is 14 words. We report three rules.

- **exact** — the cleaned answer equals the gold label.
- **first-token** — the model’s first word is a gold label word. This is the answer the model commits to, and is our headline rule.
- **set-intersection** — any word of the answer is a gold label word. This is permissive: an answer naming several parts of speech is correct as soon as one fits. We report it only to show the gap.

### 4.2 Baselines

Two reference points, both of which a model must beat to demonstrate Tigrinya knowledge:

- **Majority** — always answer the most frequent gold label (noun).
- **English-only** — an NLTK (Loper and Bird, 2002) English POS tagger applied to the English gloss alone. The prompt format is

Phrase: {Tigrinya} ({English}), and a dictionary headword’s part of speech is largely recoverable from the English word, so this baseline uses no Tigrinya whatsoever.

## 5 Results

### 5.1 POS tagging ( $n = 5,701$ )

Three models exceed both baselines: mistral-7b, gemma-7b, gemma-2b. Falcon-7b and qwen-7b fall below the majority baseline; all three seq2seq checkpoints are at or near zero.

The exact column shows how brittle these numbers are. Gemma-7B scores 0.00% exact and 79.39% first-token: it always answers in a sentence, never a bare label. A reader given only one of these columns would draw opposite conclusions.

### 5.2 Morphosyntactic labelling ( $n = 5,701$ )

Because the gold is 96.8% identical to the POS label (Section 3.2), this is effectively a second POS task.

**Only two models exceed the English-only baseline of 61.57**, by 3.02 and 1.75 points; the remaining six fall below it, and five fall below the majority baseline of 52.38.

The final column measures how often an answer contains the prompt’s own example, *preposition*, *noun*, *singular*. Gemma-7b echoes it in 97.4% of answers; because the example contains *noun*, and *noun* is the majority gold label, set-intersection scores those echoes as correct. Its apparent 89.42 is largely an artefact of copying the prompt. ByT5’s 38.55 is the same artefact at 69.7% echo, and drops to 0.00 under first-token.

### 5.3 Translation, English→Tigrinya ( $n = 5,775$ )

Tigrinya references average 1.37 whitespace tokens, so word-level BLEU-4 is near-undefined; we report chrF and character-level BLEU (Table 5).

All eight general-purpose models are at the floor. NLLB-200, for which Tigrinya (*tir\_Ethi*) is a supported target language, reaches chrF 16.05 on identical items with an identical scorer — roughly twenty times the best LLM.

Note that word-level BLEU-4 is 0.24 for NLLB-3.3B, which translates 16.24% of items exactly. **Word-level BLEU is not usable at lexicon scale**, and a paper reporting it would show a working system as scoring zero.

We verified the LLM failure is not one of output format. Few-shot prompting with eight exemplars

| Label      | Checkpoint                                             | Arch.   | Inst. |
|------------|--------------------------------------------------------|---------|-------|
| gemma-2b   | google/gemma-2b-it (Gemma Team, Google DeepMind, 2024) | causal  | yes   |
| gemma-7b   | google/gemma-7b-it (Gemma Team, Google DeepMind, 2024) | causal  | yes   |
| mistral-7b | Mistral-7B-Instruct-v0.2 (Jiang et al., 2023)          | causal  | yes   |
| qwen-7b    | Qwen1.5-7B-Chat (Bai et al., 2023)                     | causal  | yes   |
| falcon-7b  | falcon-7b-instruct (Almazrouei et al., 2023)           | causal  | yes   |
| mt5-small  | google/mt5-small (Xue et al., 2021)                    | seq2seq | no    |
| mt5-large  | google/mt5-large (Xue et al., 2021)                    | seq2seq | no    |
| byt5       | google/byt5-small (Xue et al., 2022)                   | seq2seq | no    |

Table 2: Evaluated models. Architecture and instruction-tuning status are perfectly confounded in this set.

| Model                      | exact | first-tok    | set-int |
|----------------------------|-------|--------------|---------|
| mistral-7b                 | 33.71 | <b>81.56</b> | 85.91   |
| gemma-7b                   | 0.00  | <b>79.39</b> | 82.44   |
| gemma-2b                   | 71.71 | <b>72.86</b> | 73.29   |
| falcon-7b                  | 26.94 | <b>40.64</b> | 44.45   |
| qwen-7b                    | 17.00 | <b>17.00</b> | 17.07   |
| mt5-large                  | 0.00  | 0.09         | 1.75    |
| byt5                       | 0.00  | 0.00         | 0.05    |
| mt5-small                  | 0.00  | 0.00         | 0.00    |
| <b>Majority (noun)</b>     |       | <b>53.73</b> |         |
| <b>English-only tagger</b> |       | <b>60.38</b> |         |

Table 3: POS tagging accuracy (%) under three matching rules, with both trivial baselines.

| Model                      | first-tok    | set-int | echo         |
|----------------------------|--------------|---------|--------------|
| gemma-2b                   | <b>64.59</b> | 78.02   | 1.2%         |
| gemma-7b                   | <b>63.32</b> | 89.42   | <b>97.4%</b> |
| mistral-7b                 | <b>54.92</b> | 81.09   | 3.7%         |
| falcon-7b                  | <b>39.17</b> | 76.28   | 19.5%        |
| qwen-7b                    | <b>34.33</b> | 54.45   | 0.4%         |
| mt5-small                  | 0.86         | 2.49    | 0.0%         |
| mt5-large                  | 0.11         | 0.74    | 0.0%         |
| byt5                       | 0.00         | 38.55   | 69.7%        |
| <b>Majority</b>            | <b>52.38</b> |         |              |
| <b>English-only tagger</b> | <b>61.57</b> |         |              |

Table 4: Morphosyntactic labelling accuracy (%). The final column is how often an answer contains the prompt’s own example.

and an explicit format constraint moved gemma-7b from chrF 0.16 to 0.66 with exact match unchanged at 0.00; extracting the Ge’ez span from verbose answers recovered the reference in 2 of 5,775 items. The median character-level similarity between model output and reference is 0.000: the models emit Ge’ez-shaped strings that are not the target words (ገህል for ብዙስ, ልልል as repetition).

#### 5.4 Lexical alignment

We report this task for completeness but do not draw conclusions from it: 2,208 of 5,783 alignment labels (38%) are positional fallbacks (AlignmentSource: auto) rather than annotation. Format accuracy ranges from 0.0383 (mt5-large) to 0.5484 (gemma-2b); semantic accuracy from 0.0012 (mt5-small) to 0.8876 (mistral-7b). The two measures diverge widely for individual models, which makes the task ill-posed as currently scored.

## 6 Discussion

**The matching rule is a reporting decision, not an implementation detail.** On fixed model outputs, moving from set-intersection to first-token

costs gemma-7b 3.05 points on POS and 26.10 on morphosyntax; moving to exact match costs it everything. A paper that reports one number without naming the rule has not reported a result.

**Trivial baselines are not optional.** Two baselines using no Tigrinya at all — a constant answer and an English tagger — reach 53.73 and 60.38 on POS and 52.38 and 61.57 on morphosyntax. Five of eight models fall below the stronger baseline on POS and six of eight on morphosyntax. Without those reference points, the same table reads as evidence of cross-lingual competence.

**Prompt echo is a measurable confound.** Gemma-7b reproduces the prompt’s example in 97.4% of its morphosyntax answers, and because the example contains the majority gold label, a permissive scorer rewards it. We recommend reporting echo rate wherever a prompt contains a worked example.

**Purpose-built MT still dominates at the lexicon level.** NLLB-200 checkpoints reach chrF 14.11–16.05 where the best general-purpose model

| Model            | chrF         | BLEU <sub>c</sub> | BLEU <sub>w</sub> | exact        |
|------------------|--------------|-------------------|-------------------|--------------|
| <b>NLLB-3.3B</b> | <b>16.05</b> | <b>14.23</b>      | 0.24              | <b>16.24</b> |
| NLLB-1.3B        | 15.01        | 13.29             | 0.23              | 15.00        |
| NLLB-600M        | 14.11        | 12.59             | 0.25              | 14.42        |
| byt5             | 0.81         | 0.01              | 0.00              | 0.05         |
| gemma-2b         | 0.39         | 0.01              | 0.01              | 0.07         |
| mistral-7b       | 0.26         | 0.00              | 0.00              | 0.03         |
| mt5-small        | 0.20         | 0.04              | 0.02              | 0.05         |
| gemma-7b         | 0.16         | 0.00              | 0.00              | 0.03         |
| mt5-large        | 0.13         | 0.04              | 0.01              | 0.02         |
| falcon-7b        | 0.13         | 0.00              | 0.00              | 0.03         |
| qwen-7b          | 0.04         | 0.00              | 0.00              | 0.28         |

Table 5: Translation quality. All values on a 0–100 scale.

reaches 0.81 — more than an order of magnitude on chrF. For low-resource languages, model selection dominates model size.

**What we can claim about Tigrinya competence.** Three of eight models show above-baseline POS performance. Two exceed the English-only baseline on morphosyntactic labelling, by 3.02 and 1.75 points; this margin is within the range a held-out re-run could move, and we therefore treat it as suggestive rather than established. None can translate into Tigrinya. Even NLLB reaches only 16.24% exact on a dictionary task, which indicates the difficulty is real rather than an artefact of our setup.

## 7 Relation to an earlier version of this work

An earlier version of this work, which introduced the lexicon and the four-task framework used here, has been withdrawn at the authors’ request. Its part-of-speech and morphosyntactic results were computed with the permissive matching rule described in Section 4.1, under which an answer counts as correct if any of its words matches a gold label word, and were reported without baselines; it also drew a conclusion about sequence-to-sequence architectures from a model set in which architecture and instruction-tuning are confounded.

This paper re-scores the same saved model outputs under the three matching rules of Section 4.1 and against the two baselines of Section 4.2. Measured that way, only three of the eight systems exceed both baselines on part-of-speech tagging and two on morphosyntactic labelling, in each case by a few points, while five and six respectively fall be-

low the stronger baseline; and the morphosyntactic task is not distinct from part-of-speech tagging, its gold labels being 96.8% identical.

Every figure in this paper is computed from the released lexicon, the saved generations and the scoring code listed in Section 8. No number is carried over from the withdrawn version, and the two sets of results should not be compared as alternative measurements of the same quantity.

## 8 Reproducibility

Every number in this paper is recomputed from the released artifacts by the commands in Table 6; none requires re-running a model. \$ROSTER denotes the eight evaluated models, which the scoring script takes as an explicit argument so that a reported table is reproduced over exactly that set.

The released archive also contains the raw generation for every item, stored per task and per model, so any scoring rule other than the three reported here can be applied to the same outputs without re-running the models.

## 9 Limitations

- POS labels are the source dictionary’s, not independently re-annotated. The agreement coefficients in Section 3.3 describe a 500-item subset that is alphabetical and contains no verbs, and whose adjudication is assembled from the two annotator columns rather than decided separately. They are not an estimate for the corpus; a stratified re-annotation covering verbs would be needed for that.
- Morphosyntactic features cover 185 of 5,783 gold rows.
- Lexical alignment gold is 38% positional fallback; Section 5.4 is not a result.
- Entries are dictionary headwords, not running text; performance here does not predict sentence-level performance.
- The model set cannot separate architecture from instruction-tuning. An instruction-tuned seq2seq model (flan-t5, mt0) would be needed.
- The English-only baseline shows the POS task is substantially solvable without Tigrinya. A harder task would withhold the English gloss.

| Reported in    | Command (in scripts/)                       | Artifact read                  |
|----------------|---------------------------------------------|--------------------------------|
| §3.1 counts    | build_merged_eval_set.py --dry-run --verify | Combined_POS_Lexicon.csv       |
| §3.1 5,701 set | build_sense_split_eval_set.py --verify      | *_sense_split.json             |
| §3.3 agreement | compute_iaa.py                              | *_Annotated.xlsx               |
| §5.1, §5.2     | rescore_tasks.py --models \$ROSTER          | rescored_scores.json           |
| §5.3 LLMs      | compute_bleu.py                             | translation_fidelity_bleu.json |
| §5.3 NLLB      | probe_nllb_translation.py                   | nllb_probe_*.json              |
| §5.3 few-shot  | probe_fewshot_translation.py                | fewshot_probe_*.json           |

Table 6: Each reported table, the command that regenerates it from the released artifacts, and the artifact it reads. No step requires model inference; all are computed from the saved generations.

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|     | <b>A Dataset derivation</b>                                                                                                                                                         |     |
|     | This appendix records the row-level deriva-                                                                                                                                         |     |
|     | tion summarised in Section 3.1. Every                                                                                                                                               |     |
|     | count is reproducible from the released code:                                                                                                                                       |     |
|     | <code>build_merged_eval_set.py --dry-run</code>                                                                                                                                     |     |
|     | <code>--verify</code> rebuilds the gold files from the source                                                                                                                       |     |
|     | CSV and compares them against what is re-                                                                                                                                           |     |
|     | leased, and <code>build_sense_split_eval_set.py</code>                                                                                                                              |     |
|     | <code>--verify</code> does the same for the 5,701-item set.                                                                                                                         |     |
|     | <b>Source and provenance.</b> The digitised file                                                                                                                                    |     |
|     | <code>Combined_POS_Lexicon.csv</code> holds 7,234 rows:                                                                                                                             |     |
|     | 3,587 English→Tigrinya entries followed by 3,647                                                                                                                                    |     |
|     | Tigrinya→English entries. It is a flattening of the                                                                                                                                 |     |
|     | working annotation workbook, whose eleven part-                                                                                                                                     |     |
|     | of-speech sheets hold the same 7,234 rows with                                                                                                                                      |     |
|     | the same per-category counts (3,929 noun, 1,784                                                                                                                                     |     |
|     | verb, 993 adjective, 194 adverb, and so on). The                                                                                                                                    |     |
|     | flattening lifts the inline gender and number mark-                                                                                                                                 |     |
|     | ers that the workbook keeps inside the Tigrinya                                                                                                                                     |     |
|     | field — (nf) ወኅሰት, ከሳሰት — out into the label                                                                                                                                        |     |
|     | column, which is the origin of the finer categories                                                                                                                                 |     |
|     | (nm) noun masculine, (nf) noun feminine                                                                                                                                             |     |
|     | and (npl) noun plural; the two files differ on                                                                                                                                      |     |
|     | exactly the 247 rows carrying such a marker. The                                                                                                                                    |     |

direction swap is present in the workbook as well, in 3,635 of its 7,234 rows (50.2%), so it originates in digitisation rather than in the export.

**7,234 counts rows, not independent annotations.**

Of the 3,646 unique reverse-direction pairs, 1,448 (39.7%) also occur in the forward block.

**From 7,234 to 5,783.** Merging the two directions deduplicates on the (English, Tigrinya, features) triple rather than on the pair alone, so that distinct senses of one form survive: *estimate* → ግምት is kept as both noun and verb, and *grocer* → በዓል ድኻ as both noun and noun, masculine. Expanding comma-separated senses gives 5,864 atomic pairs.

**From 5,783 to 5,775.** Each task script applies its own pair-level deduplication when loading the gold, dropping 7 duplicate (English, Tigrinya) pairs and 1 row with no English side.

**From 5,775 to 5,701.** The two tagging tasks were run against a gold file in which each multi-sense entry is represented by its first sense: (*a little*, ንእሽቶይ, ቁሩብ, ውሕድ) was evaluated as (*a little*, ንእሽቶይ). That intermediate file was not originally released, so the task scripts as published would evaluate 5,775 items rather than the 5,701 behind Sections 5.1 and 5.2. We reconstruct it from the released gold and verify the result row-for-row against the saved model outputs: 5,701 items, no row missing, none extra, and no label mismatch. The reconstructed file is released alongside the gold.